

## Chi squared test for independence



1

- a  $H_0$ : Gender and Test scores are independent.  $df = 2$
- b  $H_0$ : Gender and Job occupation are independent.  $df = 3$
- c  $H_0$ : Sport and Age are independent.  $df = 10$

2

- a  $H_0$ : The preferred sport is independent of age.  
 $H_1$ : The preferred sport is not independent of age.
- b  $df = 3$
- c  $\chi^2 = 12.15$
- d  $p = 0.00689$
- e  $p < \alpha$ , therefore we reject the null hypothesis  $H_0$ . It appears that preferred sport is not independent of age.

3

- a  $H_0$ : Gender and favourite movie genre are independent.  
 $H_1$ : Favourite movie genre is not independent of gender.
- b  $df = 2$
- c  $\chi^2 = 31.37$
- d  $\chi^2 > 11.34$ , therefore we reject the null hypothesis  $H_0$ . It appears that favourite movie genre is not independent of gender.

4

- a  $H_0$ : The level of window cleanliness is independent of the cleaning product used.  
 $H_1$ : The level of window cleanliness is not independent of the cleaning product used.
- b  $df = 6$
- c  $\chi^2 = 2.32$
- d  $p = 0.89$
- e  $p > \alpha$ , therefore we do not reject the null hypothesis  $H_0$ . It appears that window cleanliness is independent of the cleaning product used.

5

- a  $H_0$ : Camera technology preference is independent of the client's age.  
 $H_1$ : Camera technology preference is not independent of the client's age.
- b  $df = 9$
- c 40 people
- d  $\chi^2 = 57.76$

- e  $57.76 > 16.92$ , therefore we reject the null hypothesis  $H_0$ . It appears that camera technology is dependent on the age of the client.

6

- a  $H_0$ : Blood pressure level is independent of the amount of daily exercise.  
 $H_1$ : Blood pressure level is not independent of the amount of daily exercise.

b

|            | < 30   | 30 – 60 | > 60   | Total |
|------------|--------|---------|--------|-------|
| Low bp     | 12.526 | 20.632  | 15.842 | 49    |
| Average bp | 11.248 | 18.526  | 14.226 | 44    |
| High bp    | 10.226 | 16.842  | 12.932 | 40    |
| Total      | 34     | 56      | 43     | 133   |

- c  $df = 4$   
d  $\chi^2 = 58.67$   
e  $p = 0.00$   
f  $p < \alpha$ , therefore we reject the null hypothesis  $H_0$ . It appears that blood pressure level is not independent on the amount of daily exercise.

7

- $H_0$ : Voting preference is independent of gender.  
 $H_1$ : Voting preference is not independent of gender.  
The significance level  $\alpha = 0.05$ .
- $df = 2, \chi^2 = 16.20, p = 0.00$
- $p < \alpha$  therefore we reject the null hypothesis. It appears that voting preference is not independent of gender.

8

- a  $H_0$ : Buying a phone with internet access is independent of age.  
 $H_1$ : Buying a phone with internet access is not independent of age.

b

|                    | 0 – < 12 | 12 – 18 | Over 18 | Total |
|--------------------|----------|---------|---------|-------|
| Internet access    | 5        | 46      | 86      | 137   |
| No internet access | 48       | 7       | 5       | 60    |
| Total              | 53       | 53      | 91      | 197   |

- c  $df = 2$   
d  $\chi^2 = 124.63$   
e  $124.63 > 9.21$ , therefore we reject the null hypothesis  $H_0$ . It appears that buying a phone with internet access is not independent on the age of the person.